

Problem Solution-Fit

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Problem-Solution Fit :

1. CUSTOMER SEGMENT(S) [CS] i. Residents living in flood-prone areas ii. Disaster Management Authorities iii. Farmers iv. Local Government Agencies v. Environmental Researchers	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited internet access in some regions. • Lack of technical knowledge. • No access to expensive forecasting systems. • Dependence on publicly available weather information.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Existing Solutions • Government weather alerts • Manual weather monitoring • News broadcasts Pros • Easily accessible • Official information Cons • Generalized information • Delayed updates • Limited prediction capability
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Difficulty predicting floods in advance. • Heavy rainfall causes unexpected flooding. • Lack of quick flood warning systems. • Frequent during monsoon and heavy rainfall seasons.	9. PROBLEM ROOT / CAUSE [RC] i. Unpredictable weather conditions. ii. Heavy seasonal rainfall. iii. Climate change. iv. Lack of accessible real-time flood prediction	7. BEHAVIOR + ITS INTENSITY [BE] i. Regularly checks weather updates during rainy seasons. ii. Takes precautionary measures when flood warnings are issued. iii. High activity during heavy rainfall periods.

	systems.	
3. TRIGGERS TO ACT [TR] i. Heavy rainfall forecasts. ii. Rising river water levels. iii. Sudden weather changes. iv. Need for early flood preparedness.	10. YOUR SOLUTION [SL] Rising Waters is a Machine Learning-based flood prediction web application that analyzes weather parameters such as temperature, humidity, cloud cover, annual rainfall, seasonal rainfall, and average rainfall using a XG boost. The system provides instant predictions indicating whether there is a Flood Chance or No Flood Chance, enabling users to take preventive measures and improve disaster preparedness.	8. CHANNELS OF BEHAVIOR [CH] ONLINE i. Flood Prediction Web Application ii. Weather websites iii. Government portals OFFLINE • Television • Radio • Local disaster management announcements
4. EMOTIONS BEFORE / AFTER [EM] Before • Worried • Uncertain • Anxious After • Informed • Prepared • Confident		